

Abstract Algebra - Homework 2

Simon Gustafsson

Problem 1

Let $|G| = p^\alpha m$, where p is a prime, $\alpha \geq 1$ and $p \nmid m$. By definition of a p -Sylow subgroup, we have $|P| = p^\alpha$. Since $N \triangleleft G$, Lagrange's theorem implies $|N| \mid |G|$. Thus, we can write:

$$|N| = p^\beta d,$$

with $\beta \in [1, \alpha]$ and $d \mid m$, where $p \nmid d$. Consider the subgroup $N \cap P$. Since $N \cap P \leq P$, again applying Lagrange's theorem gives $|N \cap P| \mid |P| = p^\alpha$. Therefore, we must have:

$$|N \cap P| = p^\gamma,$$

where $\gamma \leq \beta$, hence $N \cap P$ must be a p -subgroup of N .

Consider the subgroup NP which contains N and P . Since $N \triangleleft G$ and $P \leq G$ we may apply the following formula,

$$|NP| = \frac{|N||P|}{|N \cap P|} = \frac{p^{\alpha+\beta}d}{p^\gamma} = p^{\alpha+\beta-\gamma}d$$

Since P is a p -Sylow subgroup of G it must also be a p -Sylow subgroup in any subgroup of G containing it. Therefore, its index in NP must be relatively prime to p .

$$[NP : P] = \frac{|NP|}{|P|} = \frac{p^{\alpha+\beta-\gamma}d}{p^\alpha} = p^{\beta-\gamma}d \implies p^{\beta-\gamma} = 1 \implies \beta = \gamma.$$

This shows $|N \cap P| = p^\beta$, which means it is a p -Sylow subgroup of N .

The quotient group G/N inherits a finite order, $|G/N| = \frac{|G|}{|N|} = \frac{p^\alpha m}{p^\beta d} = p^{\alpha-\beta} \frac{m}{d}$. A p -Sylow subgroup must therefore be of order $p^{\alpha-\beta}$. Since we

concluded $\gamma = \beta$, $|NP| = p^\alpha d$. Moreover $|N| = p^\beta d$. This allows us to construct a candidate subgroup with the correct order, NP/N

$$|NP/N| = \frac{|NP|}{|N|} = \frac{p^\alpha d}{p^\beta d} = p^{\alpha-\beta}$$

Finally, since NP is a subgroup of G containing N , the quotient NP/N is a subgroup of G/N . Therefore, NP/N is a subgroup of order $p^{\alpha-\beta}$ in G/N , and hence is a p -Sylow subgroup of G/N .

Problem 2

Problem 3

Problem 4

Problem 5

Problem 6